

GREESHMA BABU

Senior QA Automation Engineer - Mobile, Web and POS Test Automation - AI/ML Enthusiast

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PROFESSIONAL SUMMARY

- Actively building applied AI/ML expertise in RAG systems, computer vision, and enterprise AI pipelines using IBM watsonx.data, AutoAI, and Orchestration Pipelines, backed by hands-on projects and published technical content.
- 8+ years of experience in Mobile, Web, and Point of Sale test automation using Selenium, Cucumber, Java, and Appium, improving release quality and driving Agile delivery across retail and enterprise data platforms.

CORE SKILLS

- Test Automation: Selenium, Appium, Cucumber (BDD), TestNG, Robot Framework, Sikuli, Tesseract OCR, Playwright
- QA Practices: Manual Testing, Regression Testing, API Testing, Performance Testing (NeoLoad), Accessibility Testing (AssistLab), Test Planning, Test Case Design, Defect Management
- Languages and Tools: Java, Python, SQL, Maven, Jenkins, Git, JIRA, Confluence, Report Portal, YAML, CI/CD
- AI/ML: Retrieval-Augmented Generation (RAG), FAISS Vector Search, FastAPI, Streamlit, YOLO, DeepSORT, CNN-based Image Classification, IBM watsonx.data, IBM AutoAI, IBM Orchestration Pipelines
- Platforms: Android, iOS, Windows, macOS, Ubuntu, Debian, Docker
- Methodologies: Agile, Scrum, SDLC, Sprint Planning

AI AND ML PROJECTS (SELF-DIRECTED, IN PROGRESS)

Agentic RAG Assistant for Hospital Information

- Built a Retrieval-Augmented Generation system using FAISS vector search, a FastAPI backend, and a Streamlit interface to deliver context-aware hospital Q&A.

GitHub: <https://github.com/Greeshma-babu/AGENTIC-RAG>

AI-Powered Traffic Monitoring System

- Developed a real-time vehicle detection and overspeed monitoring system using YOLO for detection and DeepSORT for tracking, with violation data logged to PostgreSQL and visualized in Streamlit.

GitHub: <https://github.com/Greeshma-babu/AI-Powered-Traffic-Monitoring-System>

Image Classification Models (Characters and Celebrities)

- Built an end-to-end image classification pipeline using CNNs to identify cartoon characters and celebrity faces, covering dataset preparation, model training, validation, and testing, resulting in 94% prediction accuracy.

GitHub: https://github.com/Greeshma-babu/AI_Projects

MCP-Based Automated Defect Verification

- Building an MCP (Model Context Protocol) server on IBM BOB, integrated with Playwright and GitHub, to auto-verify defects logged as GitHub issues.
- Automated defect triage by attaching logs and screenshots as issue comments and updating defect status.
- Reduced manual QA effort by automating defect triage, evidence collection, issue updates, and status tracking through GitHub workflows.
- Recognized through participation in the IBM watsonx Challenge, where the working prototype was demonstrated and presented to senior executives and top management.

Technical Blog: Building an Enterprise AI Pipeline with IBM watsonx.data, AutoAI and Orchestration Pipelines

- Authored a tutorial on a product-success prediction use case, demonstrating an end-to-end AI workflow from raw data to production using IBM watsonx.data, AutoAI, and Orchestration Pipelines.
- Covered automated model building and workflow orchestration to enable faster insights, lower operational complexity, and enterprise-grade governance.
- Achieved 98% prediction accuracy using IBM watsonx products, showcasing the effectiveness of automated machine learning and data-driven decision-making.

PROFESSIONAL EXPERIENCE

QA Developer - IBM India Software Labs

Feb 2025 - Present

- Served as Subject Matter Expert for IBM Cloud Pak for Data, driving end-to-end data platform integration across enterprise applications.
- Executed manual and automated test cycles, identifying and raising critical defects to strengthen functional accuracy and system reliability.
- Applied IBM BOB and Agentic AI tools (AutoAI, Model Validation, Synthetic Data Generator, SPSS Modeler, Data Cleaning) to streamline testing and boost operational efficiency.
- Authored a technical blog on AI-based solutions built with IBM products and presented findings to senior management.

Senior Analyst, Web Automation (Adobe account) - Accenture

Feb 2022 - Feb 2025

- Recognized with client awards for excellence, providing extended support during critical deployments and go-lives.
- Ran performance testing with NeoLoad and accessibility testing with AssistLab to validate platform compliance and stability.
- Led manual, automation, and relay testing for new feature releases, authoring detailed test plans, test cases, and bug reports.
- Maintained and upgraded the automation framework and configured Docker on macOS to support development-environment testing.
- Drove Agile ceremonies (daily standups, sprint retrospectives) to keep QA aligned with delivery timelines.

QA Engineer, Mobile Automation (Footlocker account) - Tata Consultancy Services

May 2018 - Feb 2022

- Received client appreciation awards for excellence during critical releases and extended support windows.
- Delivered manual and automated testing across all Footlocker Android and iOS apps, safeguarding release quality.
- Conducted root-cause and failure analysis, identifying and reporting code-level and production defects.
- Participated in Agile Scrum ceremonies, supporting consistent on-time sprint delivery.

QA Engineer, POS Automation (Footlocker account) - Tata Consultancy Services

Dec 2017 - May 2018

- Won an On-the-Spot Award for completing 48 business requirements ahead of schedule.
- Converted JIRA business requirements into Cucumber feature files while actively contributing to Agile ceremonies.
- Built a BDD automation framework for POS testing, validating GUI output against customer and product data in the database.

EDUCATION

- B.Tech, Computer Science and Engineering - Federal Institute of Science and Technology, Angamaly (2013-2017), 8.25/10 CGPA

Academic Project: AI-Powered Motion Capture System

- Developed an AI-powered motion capture system that translated real-world human gestures into a virtual 3D environment using Microsoft Kinect sensor technology.
- Integrated Microsoft Kinect with Blender to enable real-time skeletal tracking and animation of a 3D character, creating an interactive virtual simulation experience.
- Implemented body-joint recognition and gesture-mapping techniques using OpenNI/NITE and OSCeleton to accurately replicate user movements in a digital avatar.
- Received the Best Project Award at the college level for innovation and successful implementation of an Artificial Intelligence solution using Microsoft Kinect.